

Groups of unipotent rank zero

SGA 3, Exposé XVI

English draft

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About this draft

This is an unofficial English translation of Exposé XVI, “Groups of unipotent rank zero” from *Schémas en groupes* (SGA 3). The source is the recomposition edited by P. Gille and P. Polo, PDF `Expo16.pdf` (24 pages), from <https://webusers.imj-prg.fr/~patrick.polo/SGA3/>. The seminar was directed by M. Demazure and A. Grothendieck (IHÉS, 1962–1964).

Statement numbers follow that PDF. A clear typographical slip is corrected in the English and marked with a `% typo:` comment; such slips are listed in the README. A slip that would change the mathematics is left as printed and marked `% typo?:`. Scholarly proofreading remains outstanding.

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Exposé XVI. Groups of unipotent rank zero

by M. Raynaud*

1. An immersion criterion

1.1. — Examples of monomorphisms of group preschemes which are not immersions. We shall construct a prescheme S , two group S -preschemes G and H , and a group S -monomorphism $u : G \rightarrow H$ which is not an immersion. The groups G and H will be of finite presentation over S , flat over S , and G will even be smooth over S (cf. Exp. VIII, the paragraph and note (*) preceding 7.1, see also XVII App. III, 4).

First take for S the spectrum of a discrete valuation ring A of mixed characteristic and with residue characteristic equal to 2. Let t be the generic point of S and s the closed point. a) Take for G the open subgroup of the constant group $G' = (\mathbf{Z}/2\mathbf{Z})_S$ obtained by removing the point of the closed fiber $(\mathbf{Z}/2\mathbf{Z})_s$ distinct from the identity element, and take for H the group of multiplicative type $(\mu_2)_S$ of second roots of unity (Exp. I 4.4.4). In view of the choice of S , H_t is isomorphic to $(\mathbf{Z}/2\mathbf{Z})_t$, whereas H_s is a radicial group. One has an evident morphism $G' \rightarrow H$, defined by the section (-1) of H , whence a morphism $u : G \rightarrow H$, which is a monomorphism but is not an immersion (otherwise it would necessarily be an isomorphism (cf. Exp. VIII 7)). Here Exp. VIII 7.9 does not apply, since ${}_2G = G$ is not finite over S .

b) Denote by K the étale nonseparated group S -prescheme obtained from the unit group by “doubling” the unique point of the closed fiber. Take for H the product group $(\mu_2)_S \times_S K$. Let a (resp. b) be the unique section of μ_2 (resp. K) over S distinct from the unit section, and let $h = (a, b)$ be the corresponding section of H . The datum h defines a group S -morphism

$$u : G = (\mathbf{Z}/2\mathbf{Z})_S \rightarrow H.$$

It is clear that u is a monomorphism, and it is not an immersion. Here Exp. VIII 7.9 does not apply, since ${}_2H = H$ is not separated over S .

c) More interesting is the following example, where G is a smooth group with connected fibers (and even $G = (\mathbf{G}_a)_S$) (cf. Exp. VIII 7.10).

Take for S the spectrum of a discrete valuation ring A of equal characteristic 2, and let π be a uniformizer of A , $\mathfrak{m}$ the maximal ideal of A .

Consider the additive group $(\mathbf{G}_a)_S$, the product group $(\mathbf{G}_a \times \mathbf{G}_a)_S$ with ring $A[X, Y]$, and let G_1 be the closed subgroup with equation

$$(1) \quad X^2 + X - \pi Y = 0.$$

The group G_1 is smooth over S , its generic fiber is isomorphic to $(\mathbf{G}_a)_t$, and its closed fiber is isomorphic to the product of $(\mathbf{G}_a)_s$ with $(\mathbf{Z}/2\mathbf{Z})_s$. One may take as factor $(\mathbf{Z}/2\mathbf{Z})_s$ the subgroup N_s of $(G_1)_s$ with equation

$$(2) \quad Y = X^2 + X = 0.$$

Let N' and N'' be two group subschemes of G_1 , isomorphic to $(\mathbf{Z}/2\mathbf{Z})_S$, whose closed fibers coincide with N_s and whose generic fibers are distinct. The groups N' and N'' are therefore

* Cf. the note on page 1 of Exp. XV.

⁰xy version of 1/12/08

defined by the datum of two sections of G_1 over S , with coordinates (a', b') (resp. (a'', b'')), such that

$$(3) \quad \begin{cases} a'^2 + a' - \pi b' = a''^2 + a'' - \pi b'' = 0, \\ a' \text{ and } a'' \equiv 1 \pmod{\mathfrak{m}}, \quad a' \neq a'', \\ b' \text{ and } b'' \equiv 0 \pmod{\mathfrak{m}}. \end{cases}$$

(One may take, for example, $(1, 0)$ and $(1 + \pi^2, \pi + \pi^3)$.)

The quotient groups $G' = G_1/N'$ and $G'' = G_1/N''$ are representable (Exp. V 4.1). Let u be the canonical morphism

$$u : G_1 \longrightarrow G' \times_S G'',$$

and let H be the group subscheme of $G' \times_S G''$ equal to the schematic closure in $G' \times_S G''$ of $u_t(G_1)_t$, so that u factors through H . The kernel of u is $K = N' \cap N''$, whose generic fiber is the unit group and whose closed fiber is equal to N_s ; in particular, K is not flat over S . On the generic fiber, u_t is therefore an isomorphism of $(G_1)_t$ onto H_t . On the other hand, I say that H_s is not smooth. Indeed, if H_s were smooth, since u_s is a morphism with finite kernel and $(G_1)_s$ and H_s have the same dimension (namely 1), u_s would be a flat morphism; since G_1 and H are flat, u would be a flat morphism, and consequently $\text{Ker } u$ would be flat over S , which it is not. It is clear that the restriction of u to the identity component G of G_1 is a monomorphism (the fibers of $K \cap G$ are unit groups, so $K \cap G$ is the unit group (Exp. VIB 2.9)), but it is not an immersion (otherwise it would be an open immersion, and H_s would be smooth). Here Exp. VIII 7.9 does not apply, since ${}_2G = G$ is not finite over S .

For enthusiasts of equations, let us say that in the above example, one may take for H the closed subgroup of $(\mathbf{G}_a \times_S \mathbf{G}_a)_S$ with ring $A[V, W]/(F)$, where

$$F = (a''b' - b''a')(a''^2V - a'^2W) - (a''V - a'W)^2$$

(where a', a'', b', b'' satisfy (3)). For G , one takes the group $(\mathbf{G}_a)_S$ with ring $A[T]$, and as morphism $u : G \longrightarrow H$, the S -morphism defined by the maps

$$V \longmapsto a'(a'T + \pi T^2), \quad W \longmapsto a''(a''T + \pi T^2).$$

Remark 1.2. The preceding construction is inspired by the method of Koizumi–Shimura⁽¹⁾. It does not apply when S is the spectrum of a ring of mixed characteristic, the points of finite order of G then being “too rigid”.

1.3. — Statement of the immersion criterion.

Theorem 1.3. *Let S be a locally noetherian prescheme, G a group S -prescheme smooth over S , of finite type, having locally for the fpqc topology Cartan subgroups (Exp. XV 6.1 and 7.3 (i)), H a group S -prescheme locally of finite type, $u : G \longrightarrow H$ a group S -monomorphism. Then:*

- a) *If G has connected fibers, for u to be an immersion, (it is necessary and) it suffices that for every S -scheme S' which is the spectrum of a complete discrete valuation ring with algebraically closed residue field, and for every Cartan subgroup C of $G_{S'}$, the restriction of $u_{S'}$ to C be an immersion.*
- b) *For u to be an immersion (resp. a closed immersion), (it is necessary and) it suffices that for every S' as above and every Cartan subgroup C of the identity component $(G_{S'})^0$ of $G_{S'}$, the restriction of $u_{S'}$ to the normalizer (Exp. XI 6.11) N of C in $G_{S'}$ be an immersion (resp. a closed immersion).*

Before proving 1.3, let us state a few applications:

Corollary 1.4. *Let S be a prescheme, G a group S -prescheme smooth over S , of finite presentation over S , of unipotent rank (Exp. XV 6.1 ter) zero, and having, locally for the*

⁽¹⁾**N.D.E.** S. Koizumi & G. Shimura, *Specialization of abelian varieties*, Sci. Papers Coll. Gen. Ed. Univ. Tokyo 9 (1959), 187–211.

fpqc topology, maximal tori, H a group S -prescheme, $u : G \rightarrow H$ a group S -monomorphism. Suppose moreover either that H is of finite presentation over S , or that S is locally noetherian and H locally of finite type. Then:

- a) If G has connected fibers, u is an immersion.*
- b) If H is separated over S and if for every S' over S and every maximal torus T of $G_{S'}$, the Weyl group*

$$W = \underline{\text{Norm}}_{G_{S'}}(T) / \underline{\text{Centr}}_{G_{S'}^0}(T)$$

is representable (a condition always satisfied if G has connected fibers (Exp. XV 7.1 (iv))) and is finite over S , then u is a closed immersion.

Corollary 1.5. *Let S be a prescheme, G a group S -prescheme smooth over S , of finite presentation over S , with connected fibers, H a group S -prescheme, $u : G \rightarrow H$ a group S -monomorphism. Suppose either that H is of finite presentation over S , or that S is locally noetherian and H locally of finite type. Then:*

- a) If G is reductive, i.e. affine over S with reductive fibers (cf. Exp. XIX), u is an immersion, and a closed immersion if H is separated.*
- b) If G has affine solvable connected fibers of unipotent rank zero, u is an immersion, and a closed immersion if H is separated.*

Proof of 1.4 from 1.3.

Reduction to the case where S is noetherian. If S is locally noetherian, the reduction is immediate, the properties to be proved being local on S . In the second case, G and H are of finite presentation over S . After restricting S , we may suppose S affine. By Exp. VIB § 10, there exist a noetherian scheme S_0 , a group S_0 -prescheme G_0 , smooth over S_0 (with connected fibers in case a)), of finite type over S_0 , a group S_0 -prescheme H_0 of finite type (separated in case b)), a group S_0 -morphism

$$u_0 : G_0 \rightarrow H_0,$$

such that G, H, u are obtained from G_0, H_0, u_0 by an extension of the base $S \rightarrow S_0$. The fact that u is a monomorphism means that $\text{Ker } u = \text{unit group}$; one may therefore suppose that u_0 is a monomorphism. Since G has unipotent rank ρ_u zero and has, locally for the fpqc topology, maximal tori, the abelian rank ρ_{ab} of the fibers of G is locally constant (Exp. XV 8.18). But ρ_u and ρ_{ab} are locally constructible functions (Exp. XV 6.3 bis). A standard argument (cf. EGA IV 8.3.4) shows that one can choose S_0 and G_0 so that the unipotent rank (resp. the abelian rank) of the fibers of G_0 is zero (resp. locally constant). But then G_0 has locally maximal tori for the fpqc topology (Exp. XV 8.18), and the functor $\mathcal{T}\mathcal{M}_{G_0}$ of maximal tori of G_0 is representable by an S_0 -scheme X_0 of finite type over S_0 (Exp. XV 8.15). Let T_0 be the “universal” maximal torus of $(G_0)_{X_0}$, X the S -prescheme $X_0 \times_{S_0} S$, $T = T_0 \times_{S_0} S$ the universal maximal torus for G . By hypothesis, in case b), the Weyl group corresponding to T is representable and finite over X . These two properties are compatible with projective limits of preschemes (Exp. VIB 10.1 iii) and EGA IV 8.10.5). One can therefore choose S_0 so that the Weyl group of T_0 is finite over X_0 . It is clear under these conditions that, to prove 1.4, one may replace S, G, u, H by S_0, G_0, u_0, H_0 , hence suppose S noetherian.

Let us use the valuative criterion provided by 1.3. We are reduced to the case where S is the spectrum of a discrete valuation ring and to the case where G^0 has a Cartan subgroup C . *Proof of 1.4 a).* We must show that the restriction of u to C is an immersion (1.3 a)), which reduces us to the case where $G = C$ has connected nilpotent fibers. Standard reductions (cf. Exp. VIII proof of 7.1) reduce us to the case where H is flat over S , u is an isomorphism on the generic fiber, and u is an isomorphism of the underlying spaces on the closed fiber. The group H then has connected fibers, hence is separated over S (Exp. VIB 5.2). Let us admit for a moment the lemma:

Lemma 1.6. *Let S be a prescheme, G a group S -prescheme smooth over S , with connected nilpotent fibers of unipotent rank zero. Then:*

- i) G is commutative.*
- ii) For every integer $n > 0$, ${}_nG$ is a group prescheme flat and quasi-finite over S , finite over S if and only if the abelian rank, or the reductive rank, of G is locally constant on S .*
- iii) For every integer $q > 0$ invertible on S , the family of subgroups ${}_qG$ is universally schematically dense in G relative to S (EGA IV 11.10.8).*

Lemma 1.6 applies to the group G , and since G has locally maximal tori, the reductive rank of the fibers of G is locally constant, so (1.6 ii)), for every integer $n > 0$, ${}_nG$ is finite over S . The fact that u is an immersion then follows from Exp. VIII 7.9.

Proof of 1.6. Let us first examine the case where S is the spectrum of a field:

Lemma 1.7. *Let k be a field of characteristic p , G an algebraic k -group, smooth, connected, nilpotent, of unipotent rank zero. Then G is a commutative group, an extension of an abelian variety A by a torus T . For every integer $n > 0$, ${}_nG$ is a finite group, étale if $(n, p) = 1$, defined by a k -algebra of rank $n^{\rho_r + 2\rho_{ab}}$. If $(q, p) = 1$, the family of subgroups ${}_qG$ ($q \in \mathbf{N}$) is schematically dense in G .*

Proof of 1.7. Let Z be the center of G . The group G/Z is affine (Exp. XII 6.1), of unipotent rank zero, smooth and connected, hence is a torus (BIBLE 4 Th. 4), but then G is commutative (Exp. XII 6.4). If T is the unique maximal torus of G (Exp. XV 3.4), it follows immediately from Chevalley's theorem (Sém. Bourbaki 1956/57, no. 145) that G is an extension of an abelian variety A by T . For every integer $n > 0$, raising to the n th power is an epimorphism in T , and one deduces an exact sequence

$$0 \longrightarrow {}_nT \longrightarrow {}_nG \longrightarrow {}_nA \longrightarrow 0.$$

A classical theorem of Weil (A. Weil: *Variétés abéliennes et courbes algébriques*, § IX Th. 33 Cor. 1) tells us that ${}_nA$ is a finite group defined by a k -algebra of rank $n^{2\rho_{ab}}$. Since ${}_nT$ is a finite group of rank n^{ρ_r} , one deduces the asserted structure of ${}_nG$.

Let then H be the smallest closed subscheme of G containing ${}_qG$ for every q . It follows from the foregoing and Exp. XV 4.6 that H is a smooth connected subgroup, hence of the same type as G . Raising to the n th power in H is an epimorphism (since ${}_qH$ is finite), so that one has the exact sequence

$$0 \longrightarrow {}_qH \longrightarrow {}_qG \longrightarrow {}_q(G/H) \longrightarrow 0.$$

It follows that ${}_q(G/H) = 0$, hence $G/H = 0$. This says that the subgroups ${}_qG$ are schematically dense in G .

Continuation of the proof of 1.6. i). To show that G is commutative, one reduces by the usual procedure to the case where S is noetherian, then to the case where S is the spectrum of a local ring with closed point s . The center of G is representable by a closed group subscheme Z of G (Exp. XI 6.11). To show that $Z = G$, it suffices to show that $Z = G$ after reduction by every power of the maximal ideal of the ring of S (for Z will then be an open subgroup of G , hence will be equal to G , since G has connected fibers). This reduces us to the case where S is local artinian.

Let q be an integer invertible on S . For every integer n , ${}_qG_s$ is an étale central subgroup of multiplicative type of G_s (1.7). Since G is smooth, ${}_qG_s$ lifts to an étale central group S -subscheme M_n of G (Exp. XV 1.2). The family of flat subgroups M_n , $n \in \mathbf{N}$, is then schematically dense in G (1.7 and EGA IV 11.10.9). Since Z is closed in G and contains all the M_n , one indeed has $Z = G$.

1.6 ii). To see that ${}_nG$ is flat and quasi-finite over S , it suffices to show that raising to the n th power in G is a flat quasi-finite morphism. Since G is flat over S , one reduces to the case where S is the spectrum of a field (EGA IV 11.3.10), and it was remarked in the proof of 1.7 that raising to the n th power was an epimorphism. Moreover, ${}_nG$ is separated over S , since G is separated over S (Exp. VIB 5.2). For ${}_nG$ to be finite over S , it is then necessary and

sufficient that the fibers of ${}_nG$ be spectra of finite algebras of locally constant rank (one sees this immediately by reducing to the case where S is the spectrum of a henselian local ring). But by 1.7, this condition is equivalent to saying that the abelian rank, or the reductive rank, of the fibers of G is locally constant.

1.6 iii). To see that the family of subgroups ${}_nG$ ($n \in \mathbf{N}$) is universally schematically dense in G , one again reduces to the case where S is noetherian. Taking account of 1.7 and 1.6 b), it then suffices to apply EGA IV 11.10.9.

Proof of 1.4 b). We have reduced to the case where S was the spectrum of a discrete valuation ring and to the case where G^0 had a Cartan subgroup C . Let $N = \underline{\text{Norm}}_G C = \underline{\text{Norm}}_G T$, where T is the unique maximal torus of C (Exp. XII 7.1 a) and b)). Since H is separated over S , it follows from 1.6 ii) and Exp. VIII 7.12 that the restriction of u to C is a closed immersion. On the other hand, to prove that u is a closed immersion, it suffices to show that this holds for $u|_N$ (1.3 b)). Since by hypothesis $W = N/C$ is representable by a finite group scheme, this will follow from the following lemma, applied to the exact sequence

$$0 \longrightarrow C \longrightarrow N \longrightarrow W \longrightarrow 1$$

(note that a proper immersion is a closed immersion).

Lemma 1.8. *Let S be a prescheme, G a group S -prescheme of finite presentation over S , an extension of a group prescheme G'' , proper and of finite presentation over S , by a group prescheme G' , of finite presentation and flat over S :*

$$1 \longrightarrow G' \longrightarrow G \longrightarrow G'' \longrightarrow 1.$$

On the other hand, let H be a group S -prescheme of finite presentation over S (or locally of finite presentation if S is locally noetherian) and $u : G \longrightarrow H$ a group S -morphism. Then if the restriction of u to G' is proper, u is proper.

Proof of 1.8. We reduce as usual to the case where S is noetherian (Exp. VIB § 10 and Exp. XV 6.2), and we can then apply the valuative criterion of properness (EGA II 7.3.8). We therefore suppose that S is the spectrum of a discrete valuation ring A , with closed point s and generic point t . Let $\bar{x} \in G(t)$ and $h \in H(S)$ be such that $u_t(\bar{x}) = h(t)$. We must show that $\bar{x}$ comes from a unique element x of $G(S)$. It even suffices to prove existence and uniqueness of x after a faithfully flat extension of the discrete valuation ring A . Now let $\bar{y}$ be the projection of $\bar{x}$ in $G''(t)$. Since G'' is proper over S , $\bar{y}$ comes from a unique element y of $G''(S)$. Let X be the inverse image of y in G . The prescheme X is faithfully flat over S (since G' is faithfully flat over S , as is the morphism $G \longrightarrow G''$ (Exp. VIB § 9)). After making a faithfully flat change of discrete valuation ring, one may suppose that X has a section ℓ over S (EGA IV 14.5.8). Replacing $\bar{x}$ by $\bar{x}\ell_t^{-1}$, one may suppose that $\bar{x} \in G'(t)$. But then existence and uniqueness of x follow from the fact that $u|_{G'}$ is proper.

Proof of 1.5. a) We shall see in Exp. XIX that if G is reductive, G has locally for the faithfully flat topology maximal tori, has a finite Weyl group and unipotent rank zero. Assertion a) follows, taking account of 1.4.

b) If G has affine fibers of unipotent rank zero, G has maximal tori locally for the fpqc topology (Exp. XV 8.18). It then suffices to apply 1.4, taking account of the fact that, since G has connected smooth affine solvable fibers, its Weyl group is the unit group (BIBLE 6 Th. 1 Cor. 3).

Proof of 1.3 a). Since the morphism u is already a monomorphism, to see that u is an immersion it suffices to show that u is proper at the points of $u(G)$ (EGA IV 15.7.1), and for this we can use the valuative criterion of local properness (EGA IV 15.7.5). We are therefore reduced to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field, with closed point s and generic point t . Since G is flat over S , standard reductions (cf. Exp. VIII proof of 7.1) allow us to reduce to the case where H is flat over S , u_t is an isomorphism and u_s is an isomorphism on the underlying spaces. To say that u is an immersion is then equivalent to saying that u is an isomorphism.

By hypothesis, the group G has locally Cartan subgroups for the fpqc topology, so we can speak of the open subset U of regular points of G (Exp. XV 7.3 i) and ii)).

Lemma 1.9. *With the preceding hypotheses, for u to be an immersion, it suffices that $u|_U$ be an immersion.*

Indeed, to say that $u|_U$ is an immersion means that there exist an open subset V of H and a closed subset F of V such that $u|_U$ factors through F and induces an isomorphism of U onto F . Since H_t , hence also V_t , is irreducible, one necessarily has $V_t = F_t$. But then F_t contains the schematic closure of V_t in V , which is equal to V , since H is flat over S . In brief, one has $V = F$. It follows that $u|_U$ is an open immersion, hence u is an open immersion (VIB 2.6).

It therefore remains for us to show that $u|_U$ is an immersion, and for this let us apply the valuative criterion of local properness. Replacing S by the spectrum of a discrete valuation ring faithfully flat over S , we must show that if h is a section of H over S whose image $h(S)$ is contained in $u(U)$, then h is the image of a section g of U over S . It suffices to show that h is contained in the image of a Cartan subgroup C of G . Indeed, by hypothesis, $u|_C$ is an immersion, so h is the image of a section g of C , which is necessarily a section of U .

Let $a \in U(s)$ (resp. $b \in U(t)$) be such that $u_s(a) = h(s)$ (resp. $u_t(b) = h(t)$). Since u_t is an isomorphism, $h(t)$ is a regular point of H_t , hence is contained in a unique Cartan subgroup D_t of H_t (Exp. XIII 3.2). Let D be the schematic closure of D_t in H .

Lemma 1.10. *i) D_s is a nilpotent algebraic group (Exp. VIB § 8).*

ii) $\dim D_s = \nu = \text{nilpotent rank of } G = \text{nilpotent rank of } (H_s)_{\text{red}}$.

iii) $h(s) \in D_s(\kappa(s))$.

i) Since H is separated (Exp. VIB 5.2), so is D . Moreover, D is flat over S , and its generic fiber is nilpotent. The fact that D_s is nilpotent then follows from Exp. VIB 8.4.

ii) Since $(H_s)_{\text{red}} \simeq G_s$, these two groups have the same nilpotent rank. The group G having locally Cartan subgroups for the fpqc topology, G_s and G_t have the same nilpotent rank ν . Finally, the group D being flat over S , one has $\dim D_s = \dim D_t = \nu$ (Exp. VIB § 4).

c) Since h_t factors through D_t , h plainly factors through D , whence iii).

This being so, the following lemma proves that $(D_s)_{\text{red}}$ is a Cartan subgroup of $(H_s)_{\text{red}} \simeq G_s$:

Lemma 1.11. *Let G be a smooth connected algebraic group defined over an algebraically closed field k , D a smooth nilpotent algebraic subgroup containing a regular element a of $G(k)$. Then D^0 is contained in a Cartan subgroup of G . If moreover $\dim D$ is equal to the nilpotent rank of G , then D is a Cartan subgroup of G (hence is connected).*

Let Z be the center of G , $G' = G/Z$, which is affine (Exp. XII 6.1), a' , D' the images of a , D in G' . It follows immediately from the correspondence between Cartan subgroups of G and Cartan subgroups of G' (Exp. XII 6.6 e)) that a' is a regular element of G' and that it suffices to prove the lemma for the pair D', G' ; this allows us to suppose G affine.

Let then s be the semisimple component of a , which belongs to $D(k)$ (BIBLE 4 Th. 3) and is regular (BIBLE 7 Th. 2 Cor. 1). By BIBLE 6 Th. 2, s centralizes D^0 , so D^0 is contained in the connected centralizer of s , which is a Cartan subgroup of G (BIBLE 7 Th. 2). If now $\dim D = \text{nilpotent rank of } G$, D^0 is therefore a Cartan subgroup of G , equal to $\text{Centr}_G(T)$, where T is the unique maximal torus of D^0 (Exp. XII 6.6). But D is nilpotent, hence centralizes T (BIBLE 6 Th. 2), so $D = D^0$.

Let us now work with the group G . The group $C_t = u_t^{-1}(D_t)$ is the unique Cartan subgroup of G_t containing b . Let C be the schematic closure of C_t in G .

By functoriality of schematic closure, $u|_C$ factors through D , so $u_s(C_s) \subset D_s$. Since u_s is an isomorphism of G_s onto $(H_s)_{\text{red}}$, $\dim C_s = \dim C_t = \nu = \dim D_s$ (1.10), and D_s is

connected (1.11), u_s provides an isomorphism of $(C_s)_{\text{red}}$ onto $(D_s)_{\text{red}}$, so $(C_s)_{\text{red}}$ is a Cartan subgroup of G_s . The following lemma shows that in fact C is a Cartan subgroup of G .

Lemma 1.12. *Let S be the spectrum of a discrete valuation ring, G a smooth group S -prescheme of finite type, C a group subscheme of G , flat over S , such that the generic fiber C_t is a Cartan subgroup of G_t and the reduced geometric closed fiber $(C_{\bar{s}})_{\text{red}}$ is a Cartan subgroup of $G_{\bar{s}}$. Then C is a Cartan subgroup of G .*

We must show that C is smooth over S , and it suffices to establish this after a faithfully flat extension of the base. The hypotheses on C imply that the nilpotent rank of the fibers of G is constant, and we can therefore speak of the open subset U of regular points of G (Exp. XV 7.3). Since $(C_{\bar{s}})_{\text{red}}$ is a Cartan subgroup of $G_{\bar{s}}$, $C_{\bar{s}}$ contains a regular point $a_{\bar{s}}$ of $G_{\bar{s}}$. But C is flat over S , so (EGA IV 14.5.8), after making a faithfully flat extension of the base, we may suppose that $a_{\bar{s}}$ is an element of $G(s)$ and lifts to a section a of $C(S)$. Since U is open and contains $a(s)$, U contains a . Let C' be the unique Cartan subgroup of G containing a (Exp. XV 7.3). One necessarily has $C_t = C'_t$, and consequently C' and C coincide with the identity component of the schematic closure of C_t in G (note that C' and C are flat over S). *End of the proof of 1.3 a).* By hypothesis, the restriction of u to the Cartan subgroup C of G is an immersion. It is clear that $h(S)$ is contained in $u(C) = D$, so h is the image of a section g of $C(S)$. Q.E.D.

Proof of 1.3 b). We shall again use the valuative criterion of local properness (EGA IV 15.5) in the case of an immersion (resp. the valuative criterion of properness (EGA II 7.3.6) in the case of a closed immersion). This reduces us to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field. Let s be the closed point and t the generic point of S . Replacing H by the schematic closure in H of $u_t(G_t)$, we may suppose that H is flat over S and u_t is an isomorphism.

Let G^0 be the connected component of G . By 1.2 a), $u|_{G^0}$ is an immersion; denote by H' the group subscheme of H which is the image of G^0 . One therefore has $H'_s = (H_s)_{\text{red}}^0$.

To verify the valuative criterion, we must show that every section h of H over S such that $h(S)$ is contained in $u(G)$ in the case of an immersion (resp. every section h of H over S in the case of a closed immersion) is the image of a section g of G over S .

Let C be a Cartan subgroup of G , $D = u(C)$ the Cartan subgroup of H' which is the image of C . The S -group $D' = \text{int}(h)D$ has connected fibers, hence has underlying space contained in that of H' ; moreover, being smooth over S , it is reduced, and consequently D' is a smooth group subscheme of H' . The fibers of D' are Cartan subgroups of the fibers of H' , so D' is a Cartan subgroup of H' , and $C' = u^{-1}(D')$ is a Cartan subgroup of G . But $\text{Transp}_G^{\text{str}}(C, C')$ is smooth and surjective over S (Exp. XII 7.1 b)). Since S is henselian with algebraically closed residue field, there exists $g \in G(S)$ such that $\text{int}(g)C = C'$. Replacing h by $u(g)^{-1}h$, we may suppose that $h(t)$ normalizes D_t (and that $h(s)$ is the image of an element of the normalizer of C_s in G_s in the case of an immersion). Let $N = \text{Norm}_G(C)$. By hypothesis, $u|_N$ is an immersion (resp. a closed immersion), so h is indeed the image of a section of N over S , which completes the proof.

2. A theorem on representability of quotients

Let us “recall” the following result:

Theorem 2.1. *Let S be a prescheme, X and Y two S -preschemes, $f : X \rightarrow Y$ an S -morphism. Suppose that one is in one of the following two cases:*

- a) *The morphism f is locally of finite presentation.*
- b) *The prescheme S is locally noetherian, and X is locally of finite type over S .*

Then the following conditions are equivalent:

i) There exists an S -prescheme X' and a factorization of f

$$f : X \xrightarrow{f'} X' \xrightarrow{f''} Y,$$

where f' is a faithfully flat S -morphism locally of finite presentation, and f'' is a monomorphism.

ii) The (first) projection

$$p_1 : X \times_Y X \longrightarrow X$$

is a flat morphism.

Moreover, if the preceding conditions hold, (X', f') is a quotient of X by the equivalence relation defined by f (for the fpqc topology), so that the factorization $f = f'' \circ f'$ of i) is unique up to isomorphism.

The proof of this delicate theorem will be found in EGA V; one may also consult the exposé of J.-P. Murre, Séminaire Bourbaki, May 1965, no. 294 Th. 2 Cor. 2, where the case of Y locally noetherian and X of finite type over Y is treated. We shall see that one can reduce to this case.

First let us make a few remarks:

a) i) $\Rightarrow$ ii) is trivial. Indeed, the first projection

$$p'_1 : X \times_{X'} X \longrightarrow X$$

factors through $X \times_Y X$:

$$p'_1 : X \times_{X'} X \xrightarrow{u} X \times_Y X \xrightarrow{p_1} X.$$

The morphism u is an isomorphism, since f'' is a monomorphism, and p'_1 is flat, since f' is flat, so p_1 is flat.

b) The assertions of 2.1 are local on Y (hence are local on S); they are also local on X , as follows easily from the fact that a flat morphism locally of finite presentation is open (EGA IV 11.3.1).

c) Under the hypotheses of 2.1 a), in view of the foregoing, we are reduced to the case where X and Y are affine and f is of finite presentation. Replacing S by Y , one may suppose X and Y of finite presentation over S . One then reduces to the case where S is noetherian by EGA IV 11.2.6.

d) Under the hypotheses of 2.1 b), one may suppose S, X, Y affine, S noetherian and X of finite type over S . Consider Y as a filtered projective limit of affine schemes Y_i of finite type over S . The schemes $X \times_{Y_i} X$ form a decreasing filtered family of closed subschemes of $X \times_S X$, whose projective limit is $X \times_Y X$. Since $X \times_S X$ is noetherian, one has $X \times_{Y_i} X = X \times_Y X$ for sufficiently large i , so $f_i : X \xrightarrow{f} Y \rightarrow Y_i$ satisfies the hypotheses of 2.1 ii) if f does. Since the equivalence relation defined by f on X coincides with that defined by f_i , it is clear that it suffices to prove ii) $\Rightarrow$ i) for f_i , which reduces us to the case where Y is of finite type over S .

Application to group preschemes

Theorem 2.2. *Let S be a prescheme, G a group S -prescheme locally of finite presentation over S , acting on an S -prescheme X . Let $\xi \in X(S)$ be a section of X over S such that the stabilizer H of ξ in G is a group subprescheme of G , flat over S . Then, if X is locally of finite type over S , or if S is locally noetherian, the quotient homogeneous space G/H is representable by an S -prescheme locally of finite presentation over S , and the S -morphism*

$$f : G \longrightarrow X, \quad g \longmapsto g \cdot \xi$$

factors as

$$\begin{array}{ccc} G & & \\ p \downarrow & \searrow f & \\ G/H & \xrightarrow{i} & X, \end{array}$$

where p is the canonical projection, which is a faithfully flat morphism locally of finite presentation, and i is a monomorphism. (To fix ideas, we have supposed that G acts on X on the left.)

Proof. The morphism f makes G an X -prescheme. By definition of the stabilizer of ξ , the morphism

$$G \times_S H \longrightarrow G \times_X G, \quad (g, h) \longmapsto (g, gh)$$

is an isomorphism. Since H is flat over S , $G \times_S H$ is flat over G , so the first projection $p_1 : G \times_S G \longrightarrow G$ is a flat morphism. Moreover, if X is locally of finite type over S , f is locally of finite presentation (EGA IV 1.4.3 v)), and otherwise S is supposed locally noetherian. It then suffices to apply 2.1 to the morphism f .

It remains to see that G/H is locally of finite presentation over S , but this follows immediately from Exp. V 9.1.

Corollary 2.3. *Let S be a prescheme, G and H two group S -preschemes, $u : G \longrightarrow H$ an S -homomorphism. Suppose G locally of finite presentation over S , and either H locally of finite type over S , or S locally noetherian. Then, if $K = \text{Ker}(u)$ is flat over S , the quotient group G/K is representable by a group S -prescheme locally of finite presentation over S , and u factors as*

$$\begin{array}{ccc} G & \xrightarrow{u} & H \\ & \searrow p & \nearrow i \\ & G/K & \end{array}$$

where p is the canonical projection and i a monomorphism.

Proof: one applies 2.2, taking $X = H$ and for ξ the unit section of H .

Corollary 2.4. *Let S be a prescheme, G a group S -prescheme of finite presentation over S , H a group S -prescheme smooth over S , with connected fibers (hence of finite presentation over S , by VIB 5.5), $i : H \longrightarrow G$ a monomorphism of S -groups, so that H is a subgroup of G .*

Suppose that $N = \underline{\text{Norm}}_G(H)$ (which is representable by a closed group subprescheme of G , of finite presentation over S (Exp. XI 6.11)) is flat over S . Then G/N is representable by an S -prescheme of finite presentation over S and quasi-projective over S .

All the assertions to be proved, except the last, are local on S . To establish them, we may therefore suppose S quasi-compact and the relative dimension of H over S constant and equal to r . Proceed as in XV § 5. For every integer $n > 0$, let $G^{(n)}$ (resp. $H^{(n)}$) be the n th normal invariant of the unit section of G (resp. of H) (EGA IV 16). The sheaf of $\mathcal{O}_S$ -modules $H^{(n)}$ is a quotient of $G^{(n)}$, and, H being smooth over S of dimension r , $H^{(n)}$ canonically defines an element ξ_n of $X_n = \text{Grass}_{\varphi(n,r)} G^{(n)}$ (EGA I 2nd ed. § 9) for a suitable integer $\varphi(n, r)$. On the other hand, G acts naturally on $G^{(n)}$ (hence also on X_n) through the representation

$$G \longrightarrow \text{Aut}_{S\text{-gr}}(G), \quad g \longmapsto \text{int}(g).$$

Since S is quasi-compact, there exists an integer m such that for $n \geq m$, one has

$$N = \underline{\text{Norm}}_G(H) = \underline{\text{Norm}}_G H^{(n)} \quad (\text{Exp. XI 6.11 b)).$$

This says that N is the stabilizer of ξ_n for large n . Representability of G/N therefore follows from 2.2. The fact that G/N is of finite presentation over S is a consequence of Exp. V 9.1. It remains to see that G/N is quasi-projective over S , and for this we shall exhibit a canonical S -ample invertible sheaf on G/N . Consider the functor $\mathcal{F} : ((\text{Sch})/S)^\circ \longrightarrow (\text{Ens})$

such that for every S -prescheme T , one has

$$\mathcal{F}(T) = \text{set of } T\text{-subgroups } H \text{ of } G_T, \text{ representable, smooth over } T, \\ \text{with connected fibers.}$$

Such a group H is of finite presentation over S (Exp. VIB 5.5), and $H \rightarrow G_T$ is a quasi-affine morphism (EGA IV 8.11.2). By effective descent of quasi-affine morphisms (SGA1 VIII 7.9), one deduces that $\mathcal{F}$ is a sheaf for the fpqc topology. Since G/N is the sheaf associated to a subsheaf of $\mathcal{F}$, one sees that there exists a canonical monomorphism $G/N \rightarrow \mathcal{F}$. There therefore exists a subgroup H' of $G \times_S (G/N)$, representable, smooth over G/N , with connected fibers, “universal” for the functor G/N . I say that the invertible sheaf $\mathcal{L} = (\det(\text{Lie } H'))^{-1}$ is S -ample. In this form, the assertion becomes local on S , and the proof is analogous to that given in (Exp. XV 5.8).

The following corollary was announced in Exp. XIV 4.8 bis.

Corollary 2.5. *Let S be a prescheme, G a group S -prescheme smooth and of finite presentation over S , with connected fibers, P a parabolic subgroup of G (Exp. XIV 4.8 bis). Then G/P is representable by an S -prescheme smooth and projective over S .*

It remains only to show (loc. cit.) that G/P is representable by a quasi-projective prescheme of finite presentation, which follows from 2.4, taking account of the fact that $P = \underline{\text{Norm}}_G(P)$ (loc. cit.), hence is flat over S .

3. Groups with flat center

Proposition 3.1. *Let S be a prescheme, G a group S -prescheme smooth and of finite presentation over S , with connected fibers. Suppose that the center Z of G (which is representable by Exp. XI 6.11) is flat over S . For every integer $n > 0$, denote by $G^{(n)}$ the locally free $\mathcal{O}_S$ -module equal to the n th normal invariant of G along the unit section, and let ρ_n be the natural “adjoint representation” of G in $\text{GL}_S(G^{(n)})$. Then:*

- a) *The quotient $G' = G/Z$ is representable by a group S -prescheme smooth, of finite presentation over S , quasi-affine, with affine connected fibers.*
- b) *The representation ρ_n factors through G' :*

$$\begin{array}{ccc} G & \xrightarrow{\rho_n} & \text{GL}_S(G^{(n)}) \\ & \searrow & \nearrow i_n \\ & G/Z & \end{array}$$

If S is quasi-compact, i_n is a monomorphism for sufficiently large n .

- c) *The functor $\mathcal{T}_{G'}$ of subtori of G' is representable by an S -prescheme smooth over S , which is a sum of a family of preschemes affine over S if S is quasi-compact.*

Proof. The assertions in a) are local on S , which allows us to suppose S quasi-compact. By Exp. XI 6.11 b), Z is equal to $\text{Ker } \rho_n$ for sufficiently large n , so the latter is flat over S . The fact that G/Z is representable and that i_n is a monomorphism therefore follows from 2.3. Since G is smooth and of finite presentation over S , the same holds for G' (Exp. VIB § 9). The group $\text{GL}_S(G^{(n)})$ is affine over S , and a monomorphism of finite presentation is quasi-affine (EGA IV 8.11.2), so G' is quasi-affine over S and has affine fibers. Since G' is smooth over S , the functor $\mathcal{T}_{G'}$ is formally smooth (Exp. XI 2.1 bis). Taking account of Exp. XI 4.6 and 4.3, the assertions in 3.1 c) will therefore follow from the following lemma:

Lemma 3.2. *Let S be a prescheme, G and H two group S -preschemes of finite presentation over S , $u : G \rightarrow H$ a group S -monomorphism, $\mathcal{T}_G$ (resp. $\mathcal{T}_H$) the functors of subtori of G (resp. of H) (cf. Exp. XV § 8). Then the map $T \mapsto u(T)$ defines a monomorphism*

(XV 8.3 c))

$$\tilde{u} : \mathcal{T}_G \longrightarrow \mathcal{T}_H$$

which is representable by a closed immersion of finite presentation.

By the usual procedure, we are reduced to the following problem: let T be a subtorus of H , $T' = G \times_T H$ its inverse image in G , $u_T : T' \longrightarrow T$ the group S -monomorphism deduced from u . One must show that the S -functor $\prod_{T/S}(T'/T)$ is representable by a closed subscheme of S , of finite presentation. But T , being a torus, is smooth over S with connected fibers, and it suffices to apply Exp. XI 6.10.

Theorem 3.3. *Let S be a prescheme, G a group S -prescheme smooth and of finite presentation over S , with connected fibers, Z the center of G . Suppose that Z is flat over S and that the unipotent rank (Exp. XV 6.1 ter) of G is equal to that of Z . Then:*

- a) *The group $G' = G/Z$ is representable, and if S is quasi-compact, the canonical morphism $i_n : G' \longrightarrow \mathrm{GL}_S(G^{(n)})$ (cf. 3.1 b)) is an immersion for large n .*
- b) *The group G' is quasi-affine over S , with affine fibers, the center of G' is the unit group, and G' has unipotent rank zero.*
- c) *The group G' has locally for the étale topology maximal tori, and these are also Cartan subgroups of G' . The functor $\mathcal{T}_{\mathcal{M}_{G'}}$ of maximal tori of G' (Exp. XV § 8) is representable by an S -prescheme smooth and affine over S .*
- d) *The group G has locally for the étale topology Cartan subgroups, and the functor $\mathcal{C}_G$ of Cartan subgroups of G is representable by an S -prescheme smooth and affine over S .*

Proof. By 3.1, the group G' is representable by an S -prescheme smooth and quasi-affine over S , with affine fibers. In view of the correspondence between Cartan subgroups of the fibers of G and of the fibers of G' (Exp. XII 6.6 e)), the hypothesis on the unipotent rank of Z implies that G' has unipotent rank zero. Using now Exp. XV 8.18, one sees that G' has locally for the étale topology maximal tori. The fact that i_n is an immersion for large n then follows from 3.1 b) and 1.4 a). This finishes proving a).

Let us now show c). Since G' has unipotent rank zero, it is clear that every maximal torus of G' is also a Cartan subgroup of G' . The functor of maximal tori of G' is representable by a subscheme, both open and closed, of the functor of subtori of G' ; moreover, it is of finite type over S (Exp. XV 8.15); it then follows from 3.1 c) that this functor is representable by an S -prescheme smooth and affine over S .

Since G' has locally for the étale topology Cartan subgroups, the same holds for G , and the functor of Cartan subgroups of G is canonically isomorphic to that of G' (Exp. XV 7.3 iv)), so c) $\Rightarrow$ d).

It remains for us to show b), and more precisely it remains to prove that the center Z' of G' is the unit group. Since Z' is representable (Exp. XI 6.11) and of finite presentation over S , it suffices to show that the fibers of Z' are reduced to the unit group (Exp. VIB 2.9), which reduces us to the case where S is the spectrum of an algebraically closed field. The center Z' is plainly contained in every Cartan subgroup of G' , hence in every maximal torus of G' by c), so is of multiplicative type. Moreover, we shall see in Exp. XVII that if G is a connected algebraic group with center Z , the center Z' of G/Z is unipotent. In the present case, Z' , being both of multiplicative type and unipotent, is reduced to the unit group (cf. Exp. XVII).

Examples of groups whose center is flat

Proposition 3.4. *Let S be a prescheme, G a group S -prescheme of finite presentation over S , smooth, with connected fibers, Z the center of G . Then Z is flat over S in the following two cases:*

- a) *The unipotent rank ρ_u (Exp. XV 6.1 ter) of the fibers of G is zero.*
- b) *i) S is reduced.*
ii) The dimension of the fibers of Z is a locally constant function on S .

iii) The unipotent rank of G is equal to the unipotent rank of Z .

Proof of 3.4 a). We shall prove at the same time the

Proposition 3.5. *Under the hypotheses of 3.4 a), suppose moreover that G has a maximal torus T , and let $C = \underline{\text{Centr}}_G(T)$ be the Cartan subgroup of G associated to T (Exp. XII 7.1). Then:*

(i) $Z \cap T$ is a subgroup of multiplicative type of G .

(ii) C is commutative and equal to $T \cdot Z$.

(iii) If the quotients $Z/Z \cap T$ and C/T are representable, they are representable by abelian preschemes (i.e. group S -preschemes smooth over S whose fibers are abelian varieties), and the canonical monomorphism $Z/Z \cap T \rightarrow C/T$ is an isomorphism.

Using the general properties of passage to the limit proved in Exp. VIB § 10 and Exp. XV 6.2, 6.3 and 6.3 bis, one reduces as usual to the case where S is noetherian (note that the assertions in 3.4 and 3.5 are local on S). We remarked in 3.1 that the hypotheses on G imply that Z is representable. To show that Z is flat over S , one then reduces by EGA 0_{III} 10.2.6 to the case where S is local artinian. But then, G being smooth, G has locally for the étale topology maximal tori (Exp. XV 8.17). After making a finite flat extension of S (which is permissible for proving that Z is flat), we may therefore suppose that G has a maximal torus T .

Let us show, in the same way, that to establish 3.5, we can reduce to the case where S is artinian.

i) Since Z is closed in G (Exp. XI 6.11), $Z \cap T$ is a closed group subscheme of T . It then follows from Exp. X 4.8 b) that $Z \cap T$ is of multiplicative type if and only if it is flat over S . As above, it suffices to establish that $Z \cap T$ is flat when S is artinian.

ii) Since G has unipotent rank zero, every Cartan subgroup C of G satisfies the hypotheses of Lemma 1.6, hence is commutative. The fact that $C = T \cdot Z$ will follow from iii).

iii) If C/T is representable, C/T is smooth over S (Exp. VIB 9), and its fibers are abelian varieties (1.7), so C/T is an abelian prescheme. To show that the canonical monomorphism

$$i : Z/Z \cap T \rightarrow C/T$$

is an isomorphism, it suffices to verify this when S is local artinian. Indeed, one will successively deduce that $Z/Z \cap T$ is flat over S (EGA 0_{III} 10.2.6), then that i is flat (EGA IV 11.3.10), then that i is an open immersion (EGA IV 17.9.1), then that i is an isomorphism (since C/T has connected fibers).

We henceforth suppose that S is local artinian with closed point s and that G has a maximal torus T . Let $C = \underline{\text{Centr}}_G(T)$. The group C is a Cartan subgroup of G (Exp. XII 7.1) and contains Z .

The algebraic group $M_s = Z_s \cap T_s$ is a subgroup of multiplicative type of G_s which is central. Since T is smooth over S , M_s lifts to a group subscheme M of T , of multiplicative type (Exp. IX 3.6 bis) and contained in the center of G (Exp. IX 3.9), hence contained in $Z \cap T$. Since S is artinian and M and T flat over S , the quotient groups $Z \cap T/M$, $Z' = Z/M$ and $A = C/T$ are representable (Exp. VIA §§ 4 and 5). By construction, $(Z \cap T/M)_s$ is the unit group, so $Z \cap T = M$ (Exp. VIB 2.9); a fortiori, it is flat over S , which proves 3.5 i).

By passage to the quotient, one has a canonical monomorphism $i : Z' \rightarrow A$, which is therefore a closed immersion (Exp. VIB 1.4.2). We have already remarked that A is an abelian scheme. It remains to show that i is an isomorphism. Indeed, this will prove 3.5 iii) and imply that Z' is flat over S , hence that Z is flat over S (as an extension of Z' by the flat group M (Exp. VIB § 9)). Since Z_s has the same abelian rank as G_s (Exp. XII 6.1), i_s is an epimorphism, hence is an isomorphism. Let q be an integer > 0 invertible on S . By the density theorem of 1.6 iii), to see that $i(Z') = A$, it suffices to show that for every integer n equal to a power of q , $i(S')$ contains ${}_n A$. Now let M_s^0 be the identity component of M_s . It is immediate by duality that raising to the q th power in M_s^0 is an epimorphism. One immediately deduces

that if m_0 is the exponent of q in the prime factorization of $\text{card}(M_s/M_s^0)$, the image of ${}_{nm_0}Z_s$ in $A_s \simeq Z_s/M_s$ contains ${}_nA_s$. There therefore exists a subgroup of multiplicative type $M_s(n)$ of Z_s whose image in A_s is ${}_nA_s$. As above, one sees that $M_s(n)$ lifts to a central subgroup of multiplicative type $M(n)$ of G . The image of $M(n)$ in A is a subgroup of multiplicative type (Exp. IX 6.8), hence necessarily equal to ${}_nA$, since this is so on the reduced fiber (Exp. IX 5.1 bis). This completes the proof of 3.4 a) and 3.5.

Proof of 3.4 b). The assertion to be proved is local on S , so we may suppose S affine with ring A . Considering A as the inductive limit of its $\mathbf{Z}$ -subalgebras of finite type, one reduces as above to the case where S is noetherian and reduced.

To show that Z is flat, one then has a valuative criterion of flatness (EGA IV 11.8.1), which allows us to reduce to the case where S is the spectrum of a complete discrete valuation ring with algebraically closed residue field, with generic point t and closed point s . Let Z' be the schematic closure in Z of Z_t . We must show that $Z' = Z$, and it even suffices to show that $Z_s = Z'_s$ (Exp. VIB 2.6). By (Exp. VIB § 4), one has the inequalities

$$\dim Z_t = \dim Z'_t = \dim Z'_s \leq \dim Z_s.$$

But by hypothesis $\dim Z_t = \dim Z_s$, so $\dim Z'_s = \dim Z_s$, and consequently Z'_s contains $(Z_s)_{\text{red}}^0$. It follows that $G'_s = G_s/Z'_s$ is affine (Exp. XII 6.1). In view of the correspondence between Cartan subgroups of G and G' (Exp. XII 6.6 e)), hypothesis 3.4 iii) implies that the unipotent rank of G'_s is zero, and consequently its Cartan subgroups are also its maximal tori. The image Z''_s of Z_s in G'_s is a central algebraic subgroup of G'_s , hence contained in every Cartan subgroup of G'_s , that is to say, in every maximal torus; Z''_s is therefore a subgroup of multiplicative type of G'_s . Under these conditions, we shall show in (Exp. XVII § 7) that there exists a finite subgroup of multiplicative type M_s of Z_s whose image in Z_s/Z'_s is Z''_s (we used this fact in the proof of 3.4 a) when Z''_s is étale). Since G is smooth over S and S is the spectrum of a complete local ring, M_s lifts to an S -subgroup of multiplicative type M of G (Exp. IX 3.6 bis and Exp. XV 1.6 b)) which is central (Exp. IX 5.6 a)). Thus M_t is contained in $Z'_t = Z_t$, and since M is flat, M is contained in Z' . The group M_s is therefore contained in Z'_s , but this implies that Z''_s is the unit group, i.e. that $Z_s = Z'_s$. This completes the proof of 3.4 b).

Example of a smooth group prescheme with connected fibers whose center is not flat Let S be the spectrum of a discrete valuation ring A , π a uniformizer of A , t the generic point of S , s the closed point. Let G be the S -group smooth and affine over S , with ring $B = A[T, T^{-1}, U]/F$, where $F = 1 - T + \pi U$, the composition law being defined by

$$(t, u), (t, u') \longmapsto (tt', \pi uu' + u + u').$$

The generic fiber G_t is therefore isomorphic to the multiplicative group $(\mathbf{G}_m)_t$, whereas the closed fiber G_s is isomorphic to the additive group $(\mathbf{G}_a)_s$. The function T is invertible in $\Gamma(G, \mathcal{O}_G)$ and defines a group S -morphism $\varphi : G \rightarrow (\mathbf{G}_m)_S$ which is an isomorphism on the generic fiber and the zero morphism on the closed fiber. The datum φ allows one to construct the semidirect product group $H = G \cdot (\mathbf{G}_a)_S$ (note that $(\mathbf{G}_m)_S$ acts on $(\mathbf{G}_a)_S$). The center Z of H is the unit group on the generic fiber and is equal to H_s on the closed fiber, hence is not flat over S .

4. Groups with affine fibers of unipotent rank zero

Theorem 4.1. *Let S be a prescheme, G a group S -prescheme smooth over S , of finite presentation, with affine connected fibers of unipotent rank (Exp. XV 6.1 ter) zero. Then:*

- a) *The center Z of G is a group of multiplicative type (and consequently is a reductive center of G (Exp. XII 4.1)).*
- b) *G satisfies conditions a) to d) of 3.3.*

c) G has locally for the étale topology maximal tori, and these are also Cartan subgroups of G . The functor $\mathcal{TM}_G$ of maximal tori of G is representable by an S -prescheme smooth and affine over S .

d) G is quasi-affine over S . Moreover, if T is a maximal torus of G , $N = \underline{\text{Norm}}_G T$ its normalizer in G , $W = N/T$ the Weyl group corresponding to T , the following conditions are equivalent:

- (i) G is affine over S .
- (ii) $G' = G/Z$ is affine over S .
- (iii) N is affine over S .
- (iv) W is affine over S .

These conditions always hold if S is locally noetherian of dimension ≤ 1 .

Proof. Since G has affine fibers (hence abelian rank zero) and unipotent rank zero, G has locally for the étale topology maximal tori (Exp. XV 8.18), and every maximal torus of G is plainly a Cartan subgroup of G , which proves the first part of c). To see that Z is of multiplicative type, one may suppose, by the foregoing, that G has a maximal torus T . Since T is also a Cartan subgroup, T contains Z (for $T = \underline{\text{Centr}}_G(T)$ by Exp. XII 7.1), and $Z = Z \cap T$ is of multiplicative type by 3.5 i). This proves a). Assertion b) is clear, taking account of a). On the other hand, the functor $\mathcal{TM}_G$ is isomorphic to the functor of Cartan subgroups of G (Exp. XII 7.1), hence is smooth and affine over S (3.3 c)), which finishes proving c). It remains to prove d).

Proof of d). Since G' is quasi-affine (3.5 b)) and Z is of multiplicative type, hence affine over S , G is quasi-affine (Exp. VIB § 9).

i) $\Leftrightarrow$ ii). If G is affine, G' is affine by Exp. IX 2.3. If G' is affine, G is affine as an extension of an affine group by an affine group (Exp. VIB § 9).

i) $\Leftrightarrow$ iii). If G is affine, N is affine since it is closed in G (Exp. XI 6.11 a)). Moreover, G/N is isomorphic to the functor $\mathcal{TM}_G$ (conjugacy of maximal tori, cf. Exp. XII 7.1 b)), hence is affine over S by c). Thus if N is affine over S , G is affine as an “extension” of an affine homogeneous space by an affine group (Exp. VIB § 9).

iii) $\Leftrightarrow$ iv). If N is affine, $W = N/T$ is affine (Exp. IX 2.3), and conversely by Exp. VIB § 9.

Moreover, N/T is representable by a group S -prescheme étale and separated over S , of finite type (Exp. XV 7.1 iv)), hence quasi-finite over S . The last assertion of d) will therefore follow from the following lemma, applied with $X = W$ and $Y = S$:

Lemma 4.2. *Let S be a locally noetherian prescheme of dimension 1, X and Y two S -preschemes locally of finite type over S , $u : X \rightarrow Y$ a quasi-finite separated S -morphism. Suppose that for every point s of S , the morphism $u_s : X_s \rightarrow Y_s$ deduced from u by the base change $\text{Spec } \kappa(s) \rightarrow S$ is finite. Then u is an affine morphism.*

The assertion to be proved is local on Y , which allows us to suppose Y (hence also X) of finite type over S . By EGA IV 8, one sees that it suffices to prove 4.2 when S is the spectrum of a local ring A . By fpqc descent, one may suppose A complete, then A reduced (EGA II 1.6.4), then A normal (Nagata’s theorem (EGA 0_{IV} 22) and Chevalley’s theorem (EGA II 6.7.1)). If A is a field, u is finite by hypothesis, hence affine. Otherwise A is a discrete valuation ring; let s be the closed point of S , t the generic point, π a uniformizer of A . Let y be a point of Y_s . Applying EGA IV 8 once again, one may replace Y by the spectrum Y' of $\mathcal{O}_{Y,y}$ and X by $X' = X \times_Y Y'$; finally, one may suppose $\mathcal{O}_{Y,y}$ complete. Since u is quasi-finite and separated, by EGA II 6.2.6, X' is the sum of two schemes X_1 and X_2 , with X_1 finite over Y' (hence affine) and X_2 such that $u(X_2)$ does not contain the closed point y of Y' . I say that under these conditions, X_2 does not meet the closed fiber X'_s . Indeed, by hypothesis, $X'_s \rightarrow Y'_s$ is finite, so the restriction of this morphism to the closed subset $(X_2)_s$ is finite, and its image in $(Y')_s$ is closed. Since this image does not contain the closed point of the local scheme $(Y')_s$, $(X_2)_s$ is empty. Since u_t is finite, the restriction to $(X_2)_t = X_2$ of the

morphism $X'_t \rightarrow Y'_t$ is finite. Moreover, one has $Y'_t = Y'_\pi$, so the open immersion $Y'_t \rightarrow Y'$ is affine. In brief, the composite morphism $X_2 \rightarrow Y'_t \rightarrow Y'$ is affine, and it indeed follows that the morphism $X' = X_1 \amalg X_2 \rightarrow Y'$ is affine.

Corollary 4.3. *Let S be a prescheme, G a group S -prescheme smooth and of finite presentation over S , with affine solvable connected fibers of unipotent rank zero. Then G is affine over S . If moreover the center of G is the unit group and S is quasi-compact, the canonical morphism $i_n : G \rightarrow \mathrm{GL}_S(G^{(n)})$ (cf. 3.1 b)) is a closed immersion for sufficiently large n .*

To prove that G is affine over S , one may suppose that G has a maximal torus T (4.1 c)). The group G having solvable fibers, the Weyl group corresponding to T is the unit group (BIBLE 6 Th. 1 Cor. 3), and condition 4.1 d) iv) is satisfied. If the center of G is the unit group, i_n is a monomorphism for sufficiently large n (3.1), hence is a closed immersion (1.5 b)).

5. Application to reductive and semisimple groups

Definition 5.1. A group S -prescheme G is said to be reductive (resp. semisimple) if G is smooth and affine over S , with reductive (resp. semisimple) fibers.

5.1.1. — Reductive groups will be systematically studied beginning with Exp. XIX. In this paragraph, we shall need the following properties, which will be proved in Exp. XIX (without using the developments of the present exposé).

- a) Let S be a prescheme, G a group S -prescheme smooth and affine over S , with connected fibers, s a point of S such that G_s is reductive (resp. semisimple). Then there exists a neighborhood U of s such that $G|_U$ is reductive (resp. semisimple).
- b) If G is reductive, G has locally for the étale topology maximal tori, and if T is a maximal torus of G , the Weyl group $W = \mathrm{Norm}_G(T)/T$ is finite over S .
- c) A reductive group has unipotent rank zero.

This being admitted, we propose to improve assertion a) above.

Theorem 5.2. *Let S be a prescheme, G a group S -prescheme smooth and of finite presentation over S , with connected fibers, s a point of S . Then:*

- (i) *If the fibers of G are affine and G_s is reductive, there exists an open neighborhood U of s such that $G|_U$ is reductive.*
- (ii) *If G_s is semisimple, there exists an open neighborhood U of s such that $G|_U$ is semisimple.*

Using Exp. XV 6.2 i) and Exp. VIB § 10, one reduces to the case where S is noetherian.

In case ii), consider the center Z of G , which is representable (Exp. XI 6.11 a)). Since G_s is semisimple, it is well known that Z_s is finite. Consequently (Exp. VIB § 4), there exists a neighborhood U of s such that Z is quasi-finite over U . For every point t of U , G_t is then affine (Exp. XII 6.1). After restricting S , we may therefore suppose that the fibers of G are affine, in case ii) as in case i).

To prove 5.2, it suffices, taking account of a), to prove that G is affine over S . Since G has affine fibers, we know that the functor of subtori of G is representable by a smooth S -prescheme (Exp. XV 8.11 and 8.9). After making an étale extension covering s , which is permissible, we may therefore suppose that there exists a subtorus T of G such that T_s is a maximal torus of G_s . But then $C = \mathrm{Centr}_G(T)$ is a smooth subgroup of G with connected fibers, containing T , such that $C_s = T_s$ (since C_s is a Cartan subgroup of G_s and G_s has unipotent rank zero by c)). It follows that $C = T$, so T is a Cartan subgroup and a maximal torus of G ; a fortiori, the unipotent rank of G is zero, and one can apply 4.1.

By 4.1 d), it suffices to show that the Weyl group W corresponding to T is affine over a neighborhood U of s . In fact, we shall see that W is even finite over a neighborhood of s . Since W is quasi-finite over S (Exp. XV 7.1 iv)), to say that W is finite over a neighborhood

U of s is equivalent to saying that the morphism $W \rightarrow S$ is proper at s (EGA IV 15.7 and EGA III 4.4.2), and to establish this one has the valuative criterion of local properness (EGA IV 15.7), which reduces us to the case where S is the spectrum of a discrete valuation ring with closed point s . But then, by the last assertion of 4.1 d), G is affine over S , and one can apply property a) recalled above to conclude that G is reductive (resp. semisimple). Using now property b), one concludes that W is indeed finite over S , which completes the proof of 5.2.

6. Applications: Extension of certain rigidity properties of tori to groups of unipotent rank zero

Proposition 6.1. *Let S be a prescheme, G a group S -prescheme smooth over S , of finite presentation, with affine connected fibers, and whose center is of multiplicative type (for example, G of unipotent rank zero, cf. 4.1 a)). Then every invariant closed subgroup K of G , of finite presentation over S , quasi-finite over S , is finite over S .*

Indeed, for every geometric point $\bar{s}$ over S , $(K_{\bar{s}})_{\text{red}}$ is a finite étale subgroup of $G_{\bar{s}}$, invariant, hence central (since $G_{\bar{s}}$ is connected). Consequently $K' = Z \cap K$ (where Z denotes the center of G) has the same underlying space as K , and it suffices to show that K' is finite over S . Now K' is a closed quasi-finite subgroup of the group of multiplicative type Z , hence is finite, as one sees immediately (locally on S , K' will be contained in ${}_nZ$ for a suitable integer n , and ${}_nZ$ is finite over S).

Corollary 6.2. *Let S and G be as above, K a group subprescheme of G , of finite presentation over S , invariant and closed in G , s a point of S . If K_s is finite (resp. is the unit group), there exists an open neighborhood U of s such that $K|_U$ is finite over S (resp. is the unit group).*

In view of the upper semicontinuity of the dimension of the fibers of K (Exp. VIB § 4), one may already suppose, after restricting S , that K is quasi-finite over S , hence finite (6.1). If moreover K_s is the unit group, one easily deduces by Nakayama's lemma that K is the unit group over $\text{Spec } \mathcal{O}_s$, hence over a neighborhood of s .

Proposition 6.3. *Let S be a prescheme, G a group S -prescheme smooth and of finite presentation over S , with affine connected fibers of unipotent rank zero, H a group S -prescheme, s a point of S , $u : G \rightarrow H$ a group S -morphism such that $\text{Ker } u_s$ is central. Suppose moreover that H is of finite presentation over S , or that S is locally noetherian and G locally of finite type over S . Then there exists an open neighborhood U of s such that, if $K = \text{Ker } u$, $K|_U$ is central, of multiplicative type. Moreover, $G' = (G|_U)/(K|_U)$ is representable, and the morphism $u' : G' \rightarrow H|_U$ deduced from u by passage to the quotient is an immersion.*

Proof. Let Z be the center of G , and $K' = K \cap Z$.

a) K' is a subgroup of multiplicative type over a neighborhood U of s . Indeed, after making an étale extension over a neighborhood U of s , which is legitimate, we may suppose that G has a maximal torus T (4.1 c)). But then $T \cap K$ is a group of multiplicative type (Exp. XV 8.3) whose fiber at s is central by hypothesis, so $T \cap K$ is central over a neighborhood of s (Exp. IX 5.6 a)), and consequently coincides with K' .

b) Let us show that $K' = K$ over a neighborhood U of s . By a), one may already suppose that K' is of multiplicative type, hence flat over S . Since $K'_s = K_s$, the natural immersion $K' \rightarrow K$ is then open over a neighborhood U of s (Exp. VIB 2.6). If $t \in U$, the image of K_t in the group $G'_t = G_t/Z_t$ is therefore a finite étale group invariant in G'_t (since K_t is invariant in G_t), hence central, hence reduced to the unit group (3.3 b)). This says that K_t is contained in Z_t , hence is equal to K'_t , whence $K = K'$ over U .

c) Representability of G' then follows from 2.3; the fact that u' is an immersion is contained in 1.3 a), taking account of 4.1 c) applied to G' .

Proposition 6.4. *Let S be a prescheme, G a group S -prescheme smooth, of finite presentation over S , with connected fibers, s a point of S such that G_s is generated by its subtori (Exp. XII 8.2) (for example, G_s affine of unipotent rank zero), H a group S -prescheme, u and $v : G \rightarrow H$ two S -homomorphisms such that $u_s = v_s$ and u_s is central. Suppose moreover that H is of finite presentation over S , or that S is locally noetherian. Then there exists a neighborhood U of s such that $u|_U = v|_U$.*

One reduces as usual to the case where S is noetherian (to study the condition “ G_s is generated by its subtori”, one uses Exp. VIB 7.4). Since G has connected fibers, to show that $u = v$ over a neighborhood U of s , it suffices to show that $u = v$ after reduction by every power of the maximal ideal of the local ring $\mathcal{O}_{S,s}$, which reduces us to the case where S is local artinian.

But then the functor of maximal subtori of G is representable by an S -scheme X , smooth and of finite type over S (Exp. XV 8.17). Let T be the maximal subtorus of G_X , universal for the functor X . The hypothesis on G_s means that the algebraic subgroup of G_s generated (Exp. VIB § 7) by the $\kappa(s)$ -morphism $f : T_s \rightarrow G_s$ (the composite of the immersion $T_s \rightarrow G_{X_s}$ and the canonical projection $G_{X_s} \rightarrow G_s$) is equal to the whole of G_s . But X_s is geometrically connected (for if T is a maximal torus of G_s and N its normalizer in G_s , X_s is isomorphic to G_s/N), and the image of T_s under f contains the unit section of G_s ; it then follows from Exp. VIB 7.4 that there exist an integer $N > 0$ and a certain S -morphism $f^N : T^N \rightarrow G$ (where $T^N = T \times_S \cdots \times_S T$ (N factors)), and f^N depends only on f and on the multiplication law of G) such that $(f^N)_s$ is surjective.

Consider the base change $X \rightarrow S$ and the restrictions of u_X and v_X to the subtorus T of G_X . By hypothesis, one has $u_{X_s} = v_{X_s}$, and $u_{X_s} : T_{X_s} \rightarrow H_{X_s}$ is a central homomorphism. It then follows from Exp. IX 5.1 that one has $u_X|_T = v_X|_T$. Making the definition of f^N explicit and taking account of the fact that u and v are homomorphisms, one immediately deduces that $uf^N = vf^N$. But the $\kappa(s)$ -morphism f_s^N is surjective, and G_s is smooth, hence reduced, so f_s^N is generically flat. Since T is smooth over S , hence flat, there exists a nonempty open subset V of T such that $f^N|_V$ is flat (EGA IV 11.3.10). The image of V is an open subset W of G (EGA IV 11.3.1), and $f^N : V \rightarrow W$ is faithfully flat, of finite presentation, hence a covering for the fpqc topology. The equality $uf^N = vf^N$ then implies $u|_W = v|_W$. Since W is schematically dense in G (EGA IV 11.10.10) and H separated over S (Exp. VIA 0.3), one immediately deduces that $u = v$.